Exact Lifespans and Round Extinction for Berger Ricci Flow

HCS-C360 source-local reconstruction

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Abstract

We give a maximal-interval classification of Ricci flow on the complete two-parameter Berger family of left-invariant metrics on $SU(2)$. A fixed Maurer–Cartan convention yields the curvature tensor and a planar metric equation. Its aspect ratio is monotone, while an anisotropy scale is exactly conserved. Two scalar charts integrate the squashed and stretched chambers: one produces an atanh lifetime and an ancient branch, and the other an arctan lifetime and a finite anisotropic backward endpoint. Every positive solution has finite forward Type-I extinction and becomes round with both metric coefficients asymptotic to four times the remaining time. The volume-normalized flow is forward complete and converges exponentially to the unique round metric on its volume leaf. Exact rational, symbolic, replay, and adversarial receipts audit the formulas without replacing the global proof. No literature-priority or arithmetic-target claim is made.

Revision certificate. Round one exact lifespan and Type I extinction closure.

1 Convention and complete atlas

Choose left-invariant one-forms satisfying

$$d\sigma_1 = 2\sigma_2 \wedge \sigma_3, \qquad d\sigma_2 = 2\sigma_3 \wedge \sigma_1, \qquad d\sigma_3 = 2\sigma_1 \wedge \sigma_2, \tag{1}$$

and consider the positive Berger cone

$$g = A(\sigma_1^2 + \sigma_2^2) + C\sigma_3^2, \qquad A, C > 0. \tag{2}$$

Changing every sign in (1) changes orientation but none of the curvature or flow formulas below. Put $r = C/A$.

Theorem 1 (Berger-flow atlas)**.** *The cone (2) is invariant under unnormalized Ricci flow, and*

$$A' = -8 + 4r, \qquad C' = -4r^2, \qquad r' = \frac{8r(1-r)}{A}. \tag{3}$$

Every positive solution has a finite forward endpoint T, satisfies

$$r(t) \longrightarrow 1, \quad A(t) \sim C(t) \sim 4(T-t), \qquad (T-t)R(t) \longrightarrow \frac{3}{2}, \tag{4}$$

and becomes extinct in a round Type-I singularity. The round and squashed branches are ancient. Every stretched branch has instead a finite backward endpoint at which $A \to 0$ and $C \to \infty$.

Under volume-normalized Ricci flow every positive solution is forward complete, preserves A^2C, and converges exponentially to the unique round metric with that volume.

2 Curvature and reduction

Let X_i be dual to σ_i and let $e_1 = X_1/\sqrt{A}$, $e_2 = X_2/\sqrt{A}$, $e_3 = X_3/\sqrt{C}$. The Koszul formula in this orthonormal Milnor frame gives

$$K_{12} = \frac{4A - 3C}{A^2}, \qquad K_{13} = K_{23} = \frac{C}{A^2}, \tag{5}$$

$$\mathrm{Ric}(e_1, e_1) = \mathrm{Ric}(e_2, e_2) = \frac{4}{A} - \frac{2C}{A^2}, \qquad \mathrm{Ric}(e_3, e_3) = \frac{2C}{A^2}. \tag{6}$$

Indeed, inserting the three rescaled structure constants into Koszul's identity $2\langle \nabla_X Y, Z\rangle = \langle [X,Y], Z\rangle - \langle [Y,Z], X\rangle + \langle [Z,X], Y\rangle$ and then into the curvature definition yields (5); summing the two planes through each e_i gives (6). Consequently

$$R = \frac{8A - 2C}{A^2}. \tag{7}$$

Since $\mathrm{Ric}(X_1, X_1) = A\,\mathrm{Ric}(e_1, e_1)$ and $\mathrm{Ric}(X_3, X_3) = C\,\mathrm{Ric}(e_3, e_3)$, the equation $g' = -2\,\mathrm{Ric}$ is precisely (3). Direct differentiation gives its ratio equation.

Lemma 1 (Anisotropy scale)**.** *Off the round ray, the positive quantity*

$$k = \frac{C}{\sqrt{|1-r|}} \tag{8}$$

is constant along every trajectory.

Proof. It is enough to square (8). In the chamber $r < 1$,

$$\frac{d}{dt}\frac{C^2}{1-r} = \frac{2CC'}{1-r} + \frac{C^2 r'}{(1-r)^2} = 0$$

after substitution from (3). Replacing $1-r$ by $r-1$ and changing the second sign gives the same cancellation above the round ray. □

Equation (3) now already determines the phase portrait: r increases below one and decreases above one. A nonround trajectory never crosses the round ray because its constant k would otherwise force C to vanish there.

3 Exact clocks and maximal intervals

In the squashed chamber set $u = \sqrt{1-r} \in (0,1)$. Lemma 1 and (3) become

$$A = \frac{ku}{1-u^2}, \qquad C = ku, \qquad u' = -\frac{4}{k}(1-u^2)^2. \tag{9}$$

Thus the remaining forward time is

$$T - t = \frac{k}{8}\left(\frac{u}{1-u^2} + \operatorname{atanh} u\right). \tag{10}$$

The primitive follows by differentiation. It is finite at $u = 0$ and diverges at $u = 1$. Hence the future endpoint has $u \to 0$, whereas backward time is infinite and $u \to 1$, $A \to \infty$, $C \to k$.

For the stretched chamber set $v = \sqrt{r-1} > 0$. Then

$$A = \frac{kv}{1+v^2}, \qquad C = kv, \qquad v' = -\frac{4}{k}(1+v^2)^2. \tag{11}$$

Its forward and backward remaining times are respectively

$$T - t = \frac{k}{8}\left(\frac{v}{1+v^2} + \arctan v\right), \tag{12}$$

$$t - T_- = \frac{k}{8}\left(\frac{\pi}{2} - \frac{v}{1+v^2} - \arctan v\right). \tag{13}$$

Both are finite. At T_- one has $v \to \infty$, so (11) gives $A \to 0$ and $C \to \infty$. These two chart exhaustions, together with the round solution $A = C = A_0 - 4t$, prove every maximal-interval assertion in Theorem 1.

As $z \to 0$, either bracket in (10) or (12) equals $2z + O(z^3)$. Therefore $z = 4(T-t)/k + O((T-t)^3)$, and (9) or (11) yields

$$A = 4(T-t) + O((T-t)^3), \qquad C = 4(T-t) + O((T-t)^3).$$

This proves (4). Equations (5) show $|\,\mathrm{Rm}\,| = O((T-t)^{-1})$. In fact the same substitution gives $(T-t)K_{13} \to 1/4$ and $(T-t)R \to 3/2$. Thus curvature actually diverges, and the endpoint is a genuine Type-I singularity rather than merely a degenerate metric-cone boundary.